

Distribution

Q1. Imulate 30 rolls with =RANDBETWEEN(1,6). What is the probability of rolling a 3 Exactly 5 times (Hint: Use BINOM.DIST)

Ans1.

$$= \text{BINOM.DIST}(5, 30, 1/6, \text{FALSE})$$
$$P = \binom{30}{5} \times \left(\frac{1}{6}\right)^5 \times \left(\frac{5}{6}\right)^{25}$$

- Combinations! 142,506
- Success: 0.0001286
- Failure: 0.01048

$$142,506 \times 0.0001286 \times 0.01048 = 0.1921$$

probability 19.21%

Q2. Generate 100 values in Excel using the continuous uniform distribution RAND() and plot a histogram. Describe the shape of the distribution.

Ans. In Excel file upload in drive.

Q3. A dataset has a mean of 50 and a standard deviation of 5. What percentage of values lie between 45 and 55 if the data follows a normal distribution?

Ans. A dataset: mean = 50, SD = 5. Find % of values between 45 and 55.

- $45 = 50 - 5 \rightarrow 1 \text{ SD below the mean}$
- $55 = 50 + 5 \rightarrow 1 \text{ SD above the mean}$

68% of values lie between 45 and 55.

Q4. What is the concept of standardization (z-score), and why is it important in data analysis? Explain the formula and how standardization transforms a dataset.

Ans. Standardization converts any data value into a universal unit — telling you how many standard deviations a value is away from the mean. The result is called a z-score.

The formula

$$z = \frac{x - \mu}{\sigma}$$

Quick Ex →

Dataset mean = 50, SD = 5.
Standardize $x = 60$.

$$z = \frac{60 - 50}{5} = +2$$

This means 60 is 2 standard deviations above the mean.

Why It's Important

1. **Compares different datasets** — e.g. scores from two different exams on the same scale
2. **Removes units** — height in cm vs weight in kg can now be compared

3. Powers ML algorithms — models like KNN, SVM, and linear regression perform better on standardized data because they are sensitive to scale

4. Detects outliers — any value beyond ± 3 is flagged as unusual

5. Enables normal distribution analysis — converts any normal distribution into the Standard Normal (Z) distribution, allowing use of z-tables

Q5. What is Kurtosis and their type?

Ans. Kurtosis measures the "tailedness" or "peakedness" of a distribution — how much of the data sits in the tails versus the center compared to a normal distribution.

Types of Kurtosis

1. Mesokurtic (Normal) — Kurtosis = 3 (Excess = 0)

- Same tail behavior as a **normal distribution**
- Neither too peaked nor too flat
- **Ex:** IQ scores, height of people

2. Leptokurtic (Heavy Tails) — Kurtosis > 3 (Excess > 0)

- Tall, sharp peak with fat/heavy tails
- More outliers than normal
- Data is highly concentrated around the mean but extreme values occur more often
- **Ex:** Stock market returns, financial crash data

3. Platykurtic (Light Tails) — Kurtosis < 3 (Excess < 0)

- Flat, broad peak with thin/light tails
- Fewer outliers than normal
- Data is more spread out, less concentrated
- **Example:** Uniform distribution, exam scores with a wide spread

Q6. Explain why the uniform distribution is a good model for the outcome of rolling a fair die

Ans. $P(x) = 1/n$

Where n = number of possible outcomes.

Applying to a Fair Die A fair die has 6 faces: {1, 2, 3, 4, 5, 6}

$P(x) = 1/6 \approx 0.1667 = 16.67\%$ for each face

Every single outcome carries exactly the same probability — no outcome is favored over another.

- Fixed number of outcomes → Exactly 6 faces
- Mutually exclusive → Can't roll two numbers at once
- Equal probability → Each face = $1/6$
- Total probability → $6 \times (1/6) = 1 \checkmark$

Key Properties

- Mean = $(1 + 6) / 2 = 3.5$
- Variance = $(6^2 - 1) / 12 \approx 2.92$
- Std Dev = $\sqrt{2.92} \approx 1.71$

Q7. Use Excel to compute the probability of getting at least 8 successes in 15 trials with success probability 0.5

Ans. In Excel file upload in drive.

Q8. How does log transformation help in stabilizing variance and making data more normally distributed?

Ans. $x \rightarrow \log(x)$

Unequal variance (heteroscedasticity)

Non-normal distribution

Poor model performance

Real Life Example

Income data:

- Most people earn ₹20,000–₹50,000
- A few earn ₹10,00,000+
- This creates a skewed distribution
- After log transform $\rightarrow$ data looks much more normal and comparable.

Log transformation only works on **positive values ($x > 0$)**. Zero and negative values will give error or undefined results.

DRIVE LINK: https://drive.google.com/drive/folders/1ZlFjacMcn7A-Gi0Qmzbq18jmXQvYluqH?usp=drive_link